MANUEL JOSE MARTE

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RESEARCH INTERESTS

I study language dysfunction and recovery after stroke. My work builds naturalistic, multimodal measures of communication from movie-viewing tasks, connected speech, and real-world language use, and pairs them with machine learning, clinical NLP, and lesion-network modeling to address the full breadth of communication impairments that persist from acute presentation through the chronic phase of recovery. Related strands examine impairment and recovery in bilingual and multilingual aphasia, disparities in recovery of communication, and the use of targeted neuromodulation. Overall, my long-term goal is to reduce persistent communication disability after stroke.

Keywords: AI/machine learning; naturalistic and multimodal assessment; discourse analysis; multilingualism; neuroimaging; neuromodulation.

EDUCATION

2025 **Ph.D., Speech, Language & Hearing Sciences**
Boston University, Boston, MA
Advisors: Swathi Kiran & Einat Liebenthal
Dissertation: *A Multimodal and Naturalistic Approach to Assessment in Aphasia*

2016 **M.S., Communication Disorders**
State University of New York at New Paltz, New Paltz, NY

2014 **B.A., Speech & Hearing Science**
University at Buffalo, Buffalo, NY

GRANT SUPPORT

Current

2026–2028 NIH/NIDCD 1L30DC024225-01
Clinical Research Loan Repayment Program
Precision Prediction and Targeted Neuromodulation for Post-Stroke Aphasia Recovery
Role: Principal Investigator

2026–2027 National Aphasia Association & Academy of Aphasia
Barbara Martin Aphasia Research Grant ($10,000)
Bedside Discourse Assessment Across Hemispheric Stroke Profiles: An Acute Feasibility Study
Role: Principal Investigator

2026–present NIH/NICHD 5T32HD007414-32
Ruth L. Kirschstein Institutional National Research Service Award
Research Training in Rehabilitation for Brain Injury and Neurological Disability
PIs: Amy J. Bastian, Christopher Thompson
Role: Postdoctoral Trainee (NRSA Fellow)

Completed

2024–2025 NIH/NIDCD 3R01DC020965-02S1
Research Supplements to Promote Diversity in Health-Related Research
Naturalistic Neuroimaging for Presurgical Language Mapping
PIs: Yanmei Tie, Einat Liebenthal
Role: Diversity Supplement Recipient

2023–2024 NIH/NIDCD 5R01DC020965
Naturalistic Neuroimaging for Presurgical Language Mapping
PIs: Yanmei Tie, Einat Liebenthal
Role: Graduate Research Assistant

2020–2022 NIH/NIDCD 5T32DC013017-05
Ruth L. Kirschstein Institutional National Research Service Award
Advanced Research Training in Communication Sciences and Disorders
PI: Christopher A. Moore
Role: Predoctoral Trainee (NRSA Fellow)

RESEARCH EXPERIENCE

Department of Neurology, Johns Hopkins University School of Medicine, Baltimore, MD

2025–present **Postdoctoral Research Fellow**
SCORE (Stroke Cognitive Outcomes & REcovery) Lab, PI: Argye E. Hillis
T32 NRSA Fellow, Kennedy Krieger Institute

- Lead development of a waitlist-controlled randomized trial staging cerebellar neuromodulation by algorithmic prediction
- Administer HD-tDCS to acute stroke patients in the Emergency Department for the TESSERACT-II trial

Institute for Technology in Psychiatry, McLean Hospital, Harvard Medical School, Belmont, MA

2024–2025 **Senior Clinical Research Assistant**
The Functional Neuroimaging & Bioinformatics Lab, PI: Einat Liebenthal

- Designed a naturalistic movie-viewing paradigm combining continuous affect ratings and eye-tracking
- Recruited and tested 100+ participants with aphasia, mild cognitive impairment, and neurotypical controls across behavioral and neuroimaging sessions

Sargent College of Health & Rehabilitation Sciences, Boston University, Boston, MA

2020–2025 **Doctoral Student Research Assistant**
Center for Brain Recovery, PI: Swathi Kiran

- Served as a lead clinician on a double-blind randomized controlled trial of semantic feature-based anomia treatment in Spanish–English bilinguals with aphasia
- Designed machine learning models predicting treatment responsiveness in bilingual aphasia from dual-language proficiency, cognitive, and demographic profiles
- Created and validated an open dataset of language experience profiles in bilinguals with and without aphasia

State University of New York at New Paltz, New Paltz, NY

2015–2016 **Graduate Student Research Assistant**

Department of Communication Disorders, PIs: Dana Arthur & Inge Anema

University at Buffalo, Buffalo, NY

2013–2014 **Undergraduate Student Research Assistant**

Center for Orofacial Pain Research, PI: Richard Ohrbach

PUBLICATIONS

‡ shared first authorship. * student mentee.

Peer-Reviewed Publications

1. **Marte, M. J.**, Chaves, M.*, Kelly, L., Diaz-Carr, I., Neal, V., Stockbridge, M. D., & Hillis, A. E. (*accepted)*. Subtle aphasia above the WAB-R cutoff one year after left-hemisphere stroke. *Aphasiology*.
2. Kiran, S., Carpenter, E., Grasemann, U., Scimeca, M., **Marte, M. J.**, Russell-Meill, M., Peñaloza, C., Tripodis, Y., & Miikkulainen, R. (2026). Predicting bilingual aphasia treatment outcomes using digital twins: A double-blind randomized controlled trial. *npj Digital Medicine*. doi: 10.1038/s41746-026-02583-9
3. Chen, X.J., **Marte, M. J.**, Kiran, S., & Blanco-Elorrieta, E. (2026). Evidence for an integrated bilingual language system from discourse tasks in aphasia. *Aphasiology*. doi: 10.1080/02687038.2026.2624735
4. Russell-Meill, M.‡, **Marte, M. J.**‡, Carpenter, E.‡, & Kiran, S. (2025). Navigating the Complexity of Bilingual Aphasia: Current Insights and Future Directions. *Brain Sciences*. doi: 10.3390/brainsci15090989
5. Scimeca, M., Peñaloza, C., Carpenter, E., **Marte, M. J.**, Russell-Meill, M., & Kiran, S. (2025). The evolution of word retrieval errors during semantic feature-based therapy in bilingual aphasia. *Bilingualism: Language and Cognition*. doi: 10.1017/S1366728925100370
6. Russell-Meill, M., Carpenter, E., **Marte, M. J.**, Scimeca, M., Peñaloza, C., & Kiran, S. (2025). Measurement of cross-language and cross-domain-generalization following semantic feature-based anomia therapy in bilingual aphasia. *Neuropsychological Rehabilitation*. doi: 10.1080/09602011.2025.2522196
7. Peñaloza, C., **Marte, M. J.**, Billot, A., & Kiran, S. (2025). Cross-language interactions during sequential anomia treatment in three languages: Evidence from a trilingual person with aphasia. *Cortex*. doi: 10.1016/j.cortex.2025.05.017
8. **Marte, M. J.**, Carpenter, E., Scimeca, M., Russell-Meill, M., Peñaloza, C., Grasemann, U., Miikkulainen, R., & Kiran, S. (2025). Machine learning predictions of recovery in bilingual post-stroke aphasia: Aligning insights with clinical evidence. *Stroke*. doi: 10.1161/STROKEAHA.124.047867
9. **Marte, M. J.**‡, Russell-Meill, M.‡, Carvalho, N.‡, & Kiran, S. (2025). Charting the Course of Aphasia Recovery: Factors, Trajectories, and Outcomes. *Annual Review of Linguistics*. doi: 10.1146/annurev-linguistics-011724-121245

10. Cordella, C., **Marte, M. J.**, Liu, H., & Kiran, S. (2025). An introduction to machine learning for speech-language pathologists: Concepts, terminology, and emerging applications. *Perspectives of the ASHA Special Interest Groups: Sig 2 Neurogenic Communication Disorders*. doi: 10.1044/2024_PERSP-24-00037
11. **Marte, M. J.**, Addesso, D.[*], & Kiran, S. (2024). Association Between Social Determinants of Health and Communication Difficulties in Poststroke U.S. Hispanic and Non-Hispanic White Populations. *American Journal of Speech-Language Pathology*. doi: 10.1044/2023_AJSLP-23-00232
12. **Marte, M. J.**, Peñaloza, C., & Kiran, S. (2023). The cognate effect on lexical access in bilingual aphasia: Evidence from the Boston Naming Test. *Bilingualism: Language and Cognition*. doi: 10.1017/S1366728923000251
13. **Marte, M. J.**[‡], Carpenter, E.[‡], Falconer, I., Scimeca, M., Abdollahi, F., Peñaloza, C., & Kiran, S. (2022). LEX-BADAT: Language EXperience in Bilinguals With and Without Aphasia DATaset. *Frontiers in Psychology*. 13:875928. doi: 10.3389/fpsyg.2022.875928

Under Review

1. **Marte, M. J.**, Gillis, B., Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., Liebenthal, E. (*under review*). Real-time emotional valence judgements during movie-watching reveal impaired language-emotion coupling in aphasia.
2. **Marte, M. J.**, Chaves, M.[*], Kelly, L., Diaz-Carr, I., Neal, V., Faria, A. V., Stockbridge, M. D., & Hillis, A. E. (*under review*). Acute-phase machine learning prediction of 12-month aphasia and discourse recovery.
3. Liebenthal, E., Girard, J. M., **Marte, M. J.**, Wei, X., Tie, Y. (*under review*). A naturalistic database for understanding individual differences in affective responses to movie clips.
4. **Marte, M. J.**, Lee, S., Wang, S., Goldin, K.[*], Varkanitsa, M., Girard, J. M., Brito, R., Wei, X., Baker, J.T., Tie, Y., Kiran, S., & Liebenthal, E. (*submitted*). Assessing Spoken Discourse in Aphasia Using Multimodal Artificial Intelligence: Development and Validation Study.
5. **Marte, M. J.**, Kelly, L., Chao, R., DeWalt, C., Gentile, M., & Stockbridge, M. D. (*submitted*). Open-weight large language models match speech-language pathologists in scoring acute aphasia from picture description.

In Preparation

1. Wei, X., **Marte, M. J.**, Baker, J. T., Tie, Y., & Liebenthal, E. (*in preparation*). Language and emotion dynamically shape neural responses to naturalistic stimuli.
2. **Marte, M. J.**, Gudenkauf, J., Kelly, L., Diaz-Carr, I., Neal, V., Faria, A. V., Stockbridge, M. D., & Hillis, A. E. (*in preparation*). Language after acute bihemispheric ischemic stroke.
3. **Marte, M. J.**, Scimeca, M., & Russell-Meill, M. (*in preparation*). Bridging or widening the gap? Artificial intelligence and equity in multilingual aphasia care. Invited contribution to *Topics in Language Disorders* special issue, *Communication Care and the Need for Equity*.
4. Stockbridge, M. D., **Marte, M. J.**, & Hillis, A. E. (*in preparation*). Task performance to discriminate mild cognitive impairment from healthy aging.
5. **Marte, M. J.**, Gillis, B., Tie, Y., Kiran, S., & Liebenthal, E. (*in preparation*). Divergent visual attention strategies in aphasia and mild cognitive impairment: Evidence from naturalistic movie-viewing.
6. Stockbridge, M. D. & **Marte, M. J.** (*in preparation*). Acetylcholinesterase Inhibition is Associated with Lower Odds of Post-Stroke Aphasia in Adults without Neurodegenerative Disorders.

HONORS & AWARDS

Honors & Competitive Appointments

2026 *ASHA Pathways Program*
ASHA Academic & Research Mentoring (ARM) Network

2025–2026 *Neuroscience Scholars Program Associate*
Society for Neuroscience

2021 *NSF Research Traineeship Program–Understanding the Brain: Neurophotonics*
DGE-1633516

2014 *Center for Undergraduate Research & Creative Activities Grant*
University at Buffalo

Conference Awards & Travel Fellowships

2026 *Trainee Professional Development Award*
Society for Neuroscience — Neuroscience 2026

NIH/NIDCD Student Scholar Award
20th Annual Eleanor M. Saffran Conference

2025 *NIH/NIDCD Young Investigator Travel Fellowship Award*
Academy of Aphasia 2025 — Mentor: Brielle Stark

NIH/NIDCD RSCA Student Fellowship Award
Clinical Aphasiology Conference 2025 — Mentor: Brielle Stark

2024 *ASHA Research Mentoring-Pair Travel Award*
ASHA Convention 2024 — Mentor: Stacy Harnish

2023 *NIH/NIDCD Young Investigator Travel Fellowship Award*
Academy of Aphasia 2023 — Mentor: Marco Calabria

NIH/NIDCD RSCA Student Fellowship Award
Clinical Aphasiology Conference 2023 — Mentor: Heather Dial

2022 *NIH/NIDCD Student Scholar Award*
16th Annual Eleanor M. Saffran Conference

INVITED TALKS

2025 **Marte, M. J.** Context matters: From individual differences to naturalistic communication in post-stroke aphasia. *SLHS Colloquium*, University of Arizona, Tucson, AZ, USA.

2024 **Marte, M. J.** Two languages, one brain: Understanding bilingual aphasia; Dos idiomas, un cerebro: Comprensión de la afasia bilingüe. *Charlas Científicas*, University of Arizona, Tucson, AZ, USA.

2023 **Marte, M. J.** Predicting treatment response in bilingual aphasia: An exploratory machine learning approach. *T32 Site Visit*, Boston University, Boston, MA, USA.

2022 **Marte, M. J.**, Carpenter, E., Scimeca, M., Kiran, S. Insights into bilingual aphasia from the PROCoM clinical trial. *T32 Clinical Connections Seminar*, University of Connecticut, Storrs, CT, USA.

PEER-REVIEWED POSTERS & PRESENTATIONS

[*] student mentee.

Oral Presentations

1. Stockbridge, M. D., Meier, E. L., & **Marte, M. J.** (2026, May). Aphasia taxonomy and linguistic exceptionalism in the Information Age: Shifting cultural perspectives. *Clinical Aphasiology Conference 2026*, Pittsburgh, PA, USA. *Roundtable*.
2. **Marte, M. J.**, Gillis, B., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2024, December). Emotional processing in post-stroke aphasia: Insights from moment-by-moment valence ratings during movie-watching. *American Speech-Language-Hearing Association Convention 2024*, Seattle, WA, USA.
3. Scimeca, M.[‡], Carpenter, E.[‡], **Marte, M. J.**[‡], & Kiran, S. (2024, March) Predicting language recovery for bilingual individuals with aphasia. *Brain Injury Association of Massachusetts 2024 Conference*, Marlborough, MA, USA.
4. **Marte, M. J.**, Addesso, D.[*], & Kiran, S. (2023, May). Social determinants of health and poststroke communication difficulties in Hispanic and non-Hispanic white adults. *Clinical Aphasiology Conference 2023*, Atlantic City, NJ, USA.
5. **Marte, M. J.**, Scimeca, M., Addesso, D.[*], Abdollahi, F., & Kiran, S. (2022, November). Hispanic Stroke Care in the United States: Insights from Epidemiology, Health Disparity, and Clinical Practice. *American Speech-Language-Hearing Association Convention 2022*, New Orleans, LA, USA.

Poster Presentations

1. **Marte, M. J.**, Chaves, M.[*], Stockbridge, M. D. & Hillis, A. E. (*submitted*). Area Deprivation Modifies Twelve-Month Language Recovery Depending on Lesion Burden. *International Stroke Conference 2027*, Houston, TX, USA.
2. Gudenkauf, J., Gezer, A., Pedagarla, C., Cortes, J. C., **Marte, M. J.**, Gudenkauf, L., Ghanem, K., & Leira, E. (*submitted*). National Survey of Clinician Practices for Syphilis Screening in Acute Stroke Settings. *International Stroke Conference 2027*, Houston, TX, USA.
3. Stockbridge, M. D. & **Marte, M. J.** (*submitted*). Acetylcholinesterase Inhibition Is Protective Against Post-Stroke Aphasia In Adults Without Neurodegenerative Disorders. *International Stroke Conference 2027*, Houston, TX, USA.
4. **Marte, M. J.**, Gudenkauf, J., Kelly, L., Diaz-Carr, I., Neal, V., Faria, A. V., Stockbridge, M. D., & Hillis, A. E. (2026, November). Vascular territory, not hemispheric bilaterality, shapes language outcomes after bilateral stroke. *Society for Neuroscience Conference 2026*, Washington, DC, USA.
5. Stockbridge, M. D., Meier, E., **Marte, M. J.**, Faria, A. V., Neal, V., Diaz-Carr, I., & Hillis, A. E. (2026, October). Feasibility of Clinical Quality Bedside Multimodal Brain Imaging. *American Neurological Association 2026 Annual Meeting*, San Diego, CA, USA.
6. **Marte, M. J.**, Chaves, M.[*], Kelly, L., Diaz-Carr, I., Neal, V., Faria, A. V., Stockbridge, M. D., & Hillis, A. E. (2026, October). Predicting Language Recovery at 12 Months from Acute Stroke Data: Machine Learning Identifies Shared and Distinct Neural Predictors. *American Neurological Association 2026 Annual Meeting*, San Diego, CA, USA.
7. Zain, N., **Marte, M. J.**, Ray, S., Stevens, R. D., & Stockbridge, M. D. (2026, October). Mapping Ischemic Stroke with Speech-Language Features: A Machine Learning Analysis of Multimodal Picture Description Data. *American Neurological Association 2026 Annual Meeting*, San Diego, CA, USA.

8. **Marte, M. J.**, Chaves, M.*, Kelly, L., Diaz-Carr, I., Neal, V., Faria, A. V., Stockbridge, M. D., & Hillis, A. E. (2026, September). Predicting Aphasia and Discourse Recovery at 12 Months From Acute Stroke Data: Shared and Distinct Neural Predictors. *20th Annual Eleanor M. Saffran Conference on Cognitive Neuroscience and Rehabilitation of Communication Disorders*, Philadelphia, PA, USA.
9. **Marte, M. J.**, Lee, S., Wang, S., Goldin, K.*, Girard, J. M., Varkanitsa, M., Tie, Y., Kiran, S., Liebenthal, E. (2026, March). Assessing spoken discourse in aphasia using multimodal artificial intelligence. *39th Annual Conference on Human Sentence Processing*, Cambridge, MA, USA.
10. Wei, X., **Marte, M. J.**, Rigolo, L., Galvin, C., Golby, A., Tie, Y., & Liebenthal, E. (2025, November). Individual differences of behavior show in dynamic brain response in movie watching. *Society for Neuroscience Conference 2025*, San Diego, CA, USA.
11. **Marte, M. J.**, Gillis, B., Goldin, K.*, Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, October). Multimodal analysis of linguistic, emotional, and visual attention processing of naturalistic movie stimuli in persons with aphasia. *Academy of Aphasia 63rd Annual Meeting*, San Diego, CA, USA.
12. **Marte, M. J.**, Gillis, B., Goldin, K.*, Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, September). Multimodal signatures in aphasia: A convergent analysis of emotion, visual attention, and language during movie-viewing. *Science of Aphasia 2025*, Copenhagen, Denmark.
13. Goldin, K.*, **Marte, M. J.**, Gillis, B., Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, July). Errors in Automatic Speech Recognition Transcripts in Neurotypical Adults and Individuals with Post-Stroke Aphasia From a Naturalistic Experiment. *Neuroscience of the Everyday World Conference 2025*, Boston, MA, USA.
14. **Marte, M. J.**, Gillis, B., Goldin, K.*, Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, July). Continuous Valence Ratings During Naturalistic Movie-Viewing Reveal Heterogeneous Language-Emotion Integration in Aphasia. *Neuroscience of the Everyday World Conference 2025*, Boston, MA, USA.
15. Wei, X., **Marte, M. J.**, Gillis, B., Galvin, C., Rigolo, L., Golby, A., Tie, Y., & Liebenthal, E. (2025, June). Understanding Brain Dynamics During Movie Watching. *Organization for Human Brain Mapping 2025 Annual Meeting*, Brisbane, Australia.
16. **Marte, M. J.**, Gillis, B., Faris, R.*, Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, June). The clinical utility of naturalistic movie-viewing: Combining emotional reactivity, gaze patterns, and contextual language processing. *Organization for Human Brain Mapping 2025 Annual Meeting*, Brisbane, Australia.
17. **Marte, M. J.**, Gillis, B., Faris, R.*, Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, May). Movie-viewing reveals novel patterns of emotional and visual processing in post-stroke aphasia. *Clinical Aphasiology Conference 2025*, Albuquerque, NM, USA.
18. **Marte, M. J.**, Gillis, B., Faris, R.*, Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, March). Naturalistic movie-viewing paradigms reveal distinct patterns of cognitive-linguistic processing across clinical populations. *Cognitive Neuroscience Society 2025 Annual Meeting*, Boston, MA, USA.
19. Russell-Meill, M., Carpenter, E., **Marte, M. J.**, Scimeca, M., Peñaloza, C., & Kiran, S. (2024, December). Outcomes of Semantic Feature-Based Therapy in Bilingual Aphasia: Patterns of Generalization Across Languages. *American Speech-Language-Hearing Association Convention 2024*, Seattle, WA, USA.
20. Russell-Meill, M., Carpenter, E., **Marte, M. J.**, Scimeca, M., Peñaloza, C., & Kiran, S. (2024, October). Beyond Naming: Promoting Generalization in Bilingual Aphasia with Semantic Feature-Based Treatment. *Academy of Aphasia 62nd Annual Meeting*, Nara, Japan.

21. Galvin, C., Rigolo, L., **Marte, M. J.**, Gillis, B., Tie, Y., Kiran, S., Liebenthal, E., & Golby, A. (2024, October). Developing a movie clip database for naturalistic brain mapping. *Society for Neuroscience Conference 2024*, Chicago, IL, USA.
22. **Marte, M. J.**, Gillis, B., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2024, October). Dynamics of emotion processing in post-stroke aphasia: Insights from continuous valence ratings during naturalistic movie viewing. *Society for Neuroscience Conference 2024*, Chicago, IL, USA.
23. Gillis, B., **Marte, M. J.**, Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2024, August). Language and emotion processing dynamics in post-stroke aphasia: Continuous and holistic valence ratings of naturalistic movie-viewing. *Neuroscience of the Everyday World 2024*, Boston, MA, USA.
24. Scimeca, M., Carpenter, E., **Marte, M. J.**, Russell, M., Peñaloza, C., & Kiran, S. (2024, May) Assessing assessment in bilingual aphasia: Towards cultural-linguistic equity in clinical testing. *Clinical Aphasiology Conference 2024*, Big Island of Hawai'i, HI, USA.
25. Carpenter, E., Grasemann, U., Scimeca, M., **Marte, M. J.**, Russell, M., Peñaloza, C., Miikkulainen, R., & Kiran, S. (2024, May) A randomized controlled trial examining the predictive power of computational simulations on treatment-induced naming outcomes in Spanish-English bilinguals with aphasia. *Clinical Aphasiology Conference 2024*, Big Island of Hawai'i, HI, USA.
26. Carpenter, E., Grasemann, U., Scimeca, M., **Marte, M. J.**, Russell, M., Peñaloza, C., Miikkulainen, R., & Kiran, S. (2024, May). Simulating therapy-induced recovery in bilingual aphasia via computational modeling and machine learning approaches. *13th World Congress for Neurorehabilitation*, Vancouver, Canada.
27. Carpenter, E., Braun, E. J., **Marte, M. J.**, Yücel, M. A., Boas, D. A., & Kiran, S. (2023, October). fNIRS Functional Connectivity in Chronic Post-Stroke Aphasia: A Pilot Study. *15th Annual Meeting of the Society for the Neurobiology of Language*, Marseille, France.
28. **Marte, M. J.**, Carpenter, E., Scimeca, M., Russell, M., Falconer, I., Peñaloza, C., & Kiran, S. (2023, October). Predicting treatment responsiveness in bilingual aphasia: The role of dual language information. *Academy of Aphasia 61st Annual Meeting*, Reading, England.
29. Carpenter, E., **Marte, M. J.**, Scimeca, M., Russell, M., Falconer, I., Peñaloza, C., & Kiran, S. (2023, October). Factors influencing language impairment in bilingual aphasia. *Academy of Aphasia 61st Annual Meeting*, Reading, England.
30. Braun, E. J., Carpenter, E., Gao, Y., **Marte, M. J.**, Yücel, M. A., Boas, D. A., & Kiran, S. (2023, May). Cortical activity patterns for narrative production in individuals with and without aphasia via functional near-infrared spectroscopy (fNIRS). *Clinical Aphasiology Conference 2023*, Atlantic City, NJ, USA.
31. Scimeca, M., Peñaloza, C., Carpenter, E., **Marte, M. J.**, Falconer, I., Combs, A., Gray, T., Shamapant, S., & Kiran, S. (2022, December). Telerehabilitation for language deficits in Spanish-English individuals with post-stroke aphasia. *12th World Congress for Neurorehabilitation*, Vienna, Austria.
32. Carpenter, E., Braun, E. J., **Marte, M. J.**, Scimeca, M., Yücel, M. A., Boas, D. A., & Kiran, S. (2022, October). Neuroscience in the Everyday World: Cortical Activation during Computer-Based Conversation Tasks Using Integrated fNIRS-EEG. *14th Annual Meeting of the Society for the Neurobiology of Language*, Philadelphia, PA, USA.
33. Addesso, D.[*†], **Marte, M. J.**, Abdollahi, F., & Kiran, S. (2022, September). Social determinants of health contributing to disparities in stroke-associated variables in Hispanic populations. *16th Annual Eleanor M. Saffran Conference on Cognitive Neuroscience and Rehabilitation of Communication Disorders*, Philadelphia, PA, USA.

[†]Selected for NIH/NIDCD Student Scholar Award.

34. **Marte, M. J.**, Addesso, D.[*], Abdollahi, F., & Kiran, S. (2022, September). U.S. Hispanic ethnicity and stroke- and neurorehabilitation-associated outcomes: Disparity-driven heterogeneity. *16th Annual Eleanor M. Saffran Conference on Cognitive Neuroscience and Rehabilitation of Communication Disorders*, Philadelphia, PA, USA.
35. Scimeca, M., Carpenter, E., **Marte, M. J.**, Combs, A., & Kiran, S. (2022, September). Enabling access to post-stroke rehabilitation in Hispanic bilinguals with aphasia: Lessons learned and challenges ahead. *16th Annual Eleanor M. Saffran Conference on Cognitive Neuroscience and Rehabilitation of Communication Disorders*, Philadelphia, PA, USA.
36. **Marte, M. J.**, Billot, A., Peñaloza, C., & Kiran, S. (2022, June). Cross-linguistic effects of a semantic treatment for anomia provided in three languages: Evidence from multilingual aphasia. *International Aphasia Rehabilitation Conference 2022*, Philadelphia, PA, USA.
37. Braun, E. J., Gao, Y., Carpenter, E., **Marte, M. J.**, Scimeca, M., Abdollahi, F., Yücel, M. A., Boas, D. A., & Kiran, S. (2021, October). Use of fNIRS to investigate cortical activation for conversational responses and narrative discourse. *Society for functional Near-Infrared Spectroscopy 2021*, virtual conference.
38. **Marte, M. J.** & Kiran, S. (2021, October). Influences on cognate facilitation in healthy bilinguals and bilinguals with aphasia. *13th Annual Meeting of the Society for the Neurobiology of Language*, virtual conference.
39. **Marte, M. J.** & Anema, I. (2015, May). Training Effects on Suprasegmental Features in Oral Reading of Non-Native English Speakers. *10th International Symposium on Bilingualism*, New Brunswick, NJ, USA.
40. Oldani, P., Hogan, C., Sauvageau, A., **Marte, M. J.**, Rudz, D., Agulia, B., Hanley, S., Quinn, T., Sharma, S., & Scerra, N., Ohrbach, R. (2014, November). The effects of varying levels of repeated anodal transcranial direct current stimulation (tDCS) of the motor cortex on visuo-motor task performance. *Biomedical Engineering Society 2014 Annual Meeting*, San Antonio, TX, USA.

TEACHING & MENTORING

Workshops

2026 *Generative AI for Statistics and Programming*.
Center for Brain Recovery, Boston University
Live demos on prompting for statistical analysis and code; output verification, reproducibility, and responsible use of sensitive data

2024 *Introduction to PsychoPy for Experimental Research*.
Center for Brain Recovery, Boston University
Hands-on session with demo scripts for building psycholinguistic experiments

2023 *Machine Learning Approaches to Analyzing Brain and Health Data*.
Hariri Institute, Boston University
Live demos on designing supervised ML pipelines for medical research data

2021 *Principal Component Analysis in Aphasia Research*.
Aphasia Research Laboratory, Boston University
Didactic and applied session on dimensionality reduction of language assessments and use questionnaires

Guest Lectures

2025 *Clinical assessment and intervention in bilingual aphasia*.
SLHS 535 Bilingualism, Multiculturalism, and Language Varieties
University of Arizona

2024 *Navigating clinical and scientific training in speech-language pathology*.
SLHS 756 Right Hemisphere Syndrome, TBI, Dementia in Communicative Disabilities
San Francisco State University

2023 *Navigating clinical and scientific training in speech-language pathology*.
SLHS 756 Right Hemisphere Syndrome, TBI, Dementia in Communicative Disabilities
San Francisco State University

Assessment and intervention in bilingual adults with aphasia.
SAR SH725 Bilingual Considerations in Language Assessment and Intervention
Boston University

Topics in bilingual aphasia.
CD 690 Aphasia
Emerson College

2022 *Introduction to bilingual aphasia*.
SAR SH 548 Introduction to Language Disorders Across the Lifespan
Boston University

Topics in bilingual aphasia.
CD 690 Aphasia ($\times$ 2)
Emerson College

2021 *Topics in bilingual aphasia*.
CD 690 Aphasia
Emerson College

Co-instructor

2016 CMD 301 Speech Science
State University of New York at New Paltz
Co-taught lectures, led laboratory and discussion sections

Teaching Assistant

2015 CMD 301 Speech Science
CMD 305 Hearing Science
CMD 310 Anatomy & Physiology of the Speech and Hearing Mechanism
State University of New York at New Paltz
Led laboratory and discussion sections, facilitated review sessions, graded coursework

2014 CMD 305 Hearing Science
CMD 310 Anatomy & Physiology of the Speech and Hearing Mechanism
State University of New York at New Paltz
Led laboratory and discussion sections, facilitated review sessions, graded coursework

Tutor

2014–2015 CMD 301 Speech Science, CMD 305 Hearing Science, CMD 310 Anatomy & Physiology of the Speech and Hearing Mechanism
State University of New York at New Paltz

Research Mentorship

[UG] undergraduate. [G] graduate. [HS] high school.

2026–present Vidhi Bansal (Johns Hopkins University '29)[UG]
2026–present Abhinav Rajkumar (Baylor University '27)[UG]
2026–present Mathew Chaves (Johns Hopkins University '26)[UG] — Lesion-network prediction of discourse recovery; co-authored poster at American Neurological Association 2026
2025–present Rachel Kollarmalil (Johns Hopkins University '30)[UG] — Undergraduate Research Symposium submission
2025–present Vandie Dumaboc (Harvard University '27)[UG] — Undergraduate neuroscience thesis
2025 Kari Goldin (University of Massachusetts Amherst '26)[UG] — ASR error analysis in aphasia; co-authored posters at Academy of Aphasia 2025 and Neuroscience of the Everyday World 2025
2024–2025 Isabel Yu (Boston University '27)[UG] — Co-mentored with Michael Scimeca; bilingual aphasia eye-tracking poster co-author
2024–2025 Rowan Faris (University of Massachusetts Amherst '27)[UG] — Co-authored posters at OHBM 2025, CAC 2025, and CNS 2025
2023–2025 *Supervisor*, Conversation, Health, Art, Technology (CHAT) Group — community aphasia conversation group; supervised student clinicians and volunteer facilitators
2022–2023 Sofia Garcia Martinez (University of Illinois at Chicago '23)[G]
M.S. in Biomedical Visualization, Master's Thesis Deliverable [Sample]
2021–2023 David Addesso (Boston University '22)[UG]
UROP Mentee — Co-authored publication in *AJSLP* and presentations at Saffran Conference 2022; selected for NIH/NIDCD Student Scholar Award
2021 Hongli Zheng (Boston University '22)[UG]
UROP Mentee
2021 Emily Benson-Tyler (Port Washington High School '22)[HS]
Research in Science & Engineering Program Mentee; *Regeneron Science Talent Search*

SERVICE

Service to the Field

Ad hoc reviewer (53+ reviews): *American Journal of Speech-Language Pathology*, *Aphasiology*, *Behavioral Sciences*, *BioMedical Engineering OnLine*, *BMC Medical Informatics and Decision Making*, *BMC Neurology*, *Brain Communications*, *Clinical Linguistics & Phonetics*, *Cognitive Processing*, *Cortex*, *Diagnostics*, *Digital Health*, *Discover Neuroscience*, *International Journal of Multilingualism*, *JAMA Neurology*, *Journal of Communication Disorders*, *Journal of Speech, Language, and Hearing Research*, *Memory & Cognition*, *NeuroImage: Clinical*, *Neurological Research*, *Neurological Research and Practice*, *Neurorehabilitation and Neural Repair*, *npj Artificial Intelligence*, *npj Dementia*, *npj Digital Public Health*, *Science Progress*, *Scientific Data*, *Scientific Reports*, *Stroke*

2026 Reviewer, 2026 National Aphasia Association Grant Proposals

2025 Volunteer, Neuroscience of the Everyday World 2025

2023 Volunteer, Neuroscience of the Everyday World 2023

2022 Session Chair, BIG BRAIN WORKSHOP 2022

2021 Talk and Poster Moderator, 46th Annual Boston University Conference on Language Development

Service to the Community

2026 Judge, Youth 4 Neuro Neuroscience Research Competition
Student-led 501(c)(3) nonprofit promoting neuroscience education and career pathways for secondary-school students

2026 Reviewer, Provost's Undergraduate Research Award (PURA), Hopkins Office for Undergraduate Research

2025 Judge, Youth 4 Neuro Neuroscience Research Competition

2022 Judge, Massachusetts High School Science & Engineering Fair

Professional Memberships

Society for Neuroscience
Cognitive Neuroscience Society
ASHA SIG 2: Neurogenic Communication Disorders
ASHA SIG 14: Cultural and Linguistic Diversity

PROFESSIONAL DEVELOPMENT

2025 *New England Future Faculty Workshop*
Two-day faculty job-market workshop; application materials reviewed with personalized faculty feedback; interviewing and negotiation

2025 *Oxford Clinical Neuroimaging Course*, Wellcome Centre for Integrative Neuroimaging
Advanced MRI interpretation and applications in clinical populations; guided FSL analysis tutorials with real data

2024 *NIH Diversity Supplement Professional Development and Networking Workshop*
Grantsmanship and the transition to independent investigator

2023 *ASHA Mentoring Academic-Research Careers (MARC) Program*
Year-long mentorship on academic research careers

2022 *ASHA Minority Student Leadership Program (MSLP)*
Leadership development for students underrepresented in the discipline

2022 *ASHA Student to Empowered Professional (S.T.E.P.) Mentoring Program*
Student-to-professional mentoring

2018 *Foundations in Myofascial Release for Neck, Voice, and Swallowing Disorders*
Pain Relief Center — manual therapy for voice and swallowing

SELECTED CLINICAL EXPERIENCE

2016–2020 *Northeast Center for Rehabilitation and Brain Injury*, Lake Katrine, NY
Speech–Language Pathologist and Practicum Supervisor. Evaluation and treatment services for adults with augmentative and alternative communication needs, ventilator dependency, executive dysfunction, pragmatic deficits, short- and long-term memory deficits, disorders of consciousness, voice, lan-

guage, motor speech, auditory processing, neurogenic fluency, cognitive communication, and feeding and swallowing disorders. Mentoring and oversight of graduate-level student clinicians in this setting.

2018–2020 *Ghent Rehabilitation and Nursing Center*, Ghent, NY
Speech–Language Pathologist. Evaluation and treatment services for older adults with dementia, short- and long-term memory deficits, language, cognitive communication, and feeding and swallowing disorders.

PROFESSIONAL LICENSES AND CERTIFICATIONS

Certificate of Clinical Competence from the American Speech–Language–Hearing Association
Licensed Speech–Language Pathologist, State of Maryland

MEDIA

2026 The Scientist: *Postdoc Portrait: Manuel Marte Uses Computational Models to Predict Stroke Recovery.*
2026 Futurism at JHU Podcast: *Decoding Aphasia and Recovery From Stroke.*
2025 American Heart Association News: *Speaking more than one language can add layers to stroke recovery.*

SKILLS

Languages: English (native), Spanish (native)

Programming: Python (scikit-learn, PyTorch, pandas, spaCy, Hugging Face), R, Bash, Git, SLURM/HPC

Machine learning & statistics: Supervised learning and model interpretability; clustering and dimensionality reduction; nested cross-validation; mixed-effects and hierarchical models

NLP, speech & LLMs: Automatic speech recognition; semantic embeddings and transformer fine-tuning; open-weight LLM deployment; automated discourse and main concept analysis; CLAN/CHAT (TalkBank)

Neuroimaging & connectomics: FSL, FreeSurfer, ANTs; lesion-network mapping (ChaCo/NeMo); clinical-grade MRI processing

Experimental & clinical research: PsychoPy, CARMA, eye-tracking, HD-tDCS administration, REDCap, OnCore, IRB protocol development

Clinical: Assessment and treatment of aphasia, cognitive-communication, motor speech, AAC, and dysphagia in adult neurorehabilitation; clinical supervision

REFERENCES

Available upon request.